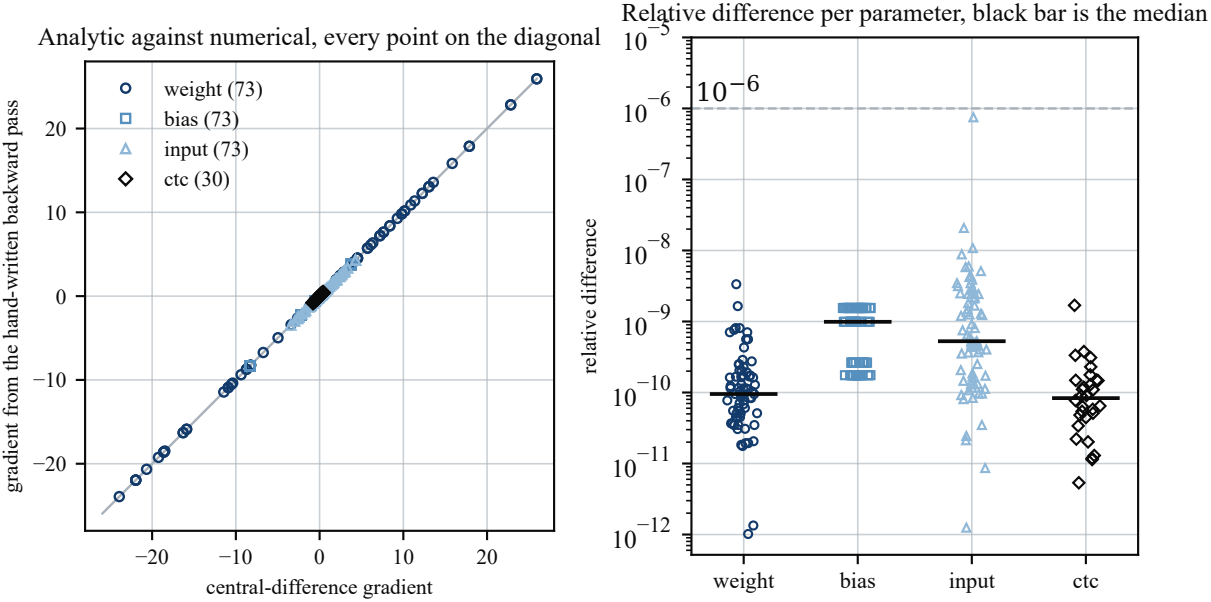
Analytic against numerical, every point on the diagonal
weight (73)
bias (73)
input (73)
ctc (30)
gradient from the hand-written backward pass
central-difference gradient
20
10
0
−10
−20
−20
−10
0
10
20
Relative difference per parameter, black bar is the median
relative difference
10^{-6}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
weight
bias
input
ctc